Card & Krueger (1994) — Reproduction

Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania · AER 84(4), 772–793

Data: `data/public.csv` (410 fast-food stores). FTE employment = full-time employees + managers + 0.5 × part-time employees. Cells show the mean with its standard error in parentheses.

Table 3 — Average employment per store before and after the New Jersey minimum-wage rise

	PA	NJ	Diff NJ–PA	NJ wage =$4.25	NJ wage $4.26–4.99	NJ wage ≥$5.00	Low–high	Mid–high
FTE employment before	23.33 (1.35)	20.44 (0.51)	-2.89 (1.44)	19.56 (0.77)	20.08 (0.84)	22.25 (1.14)	-2.69 (1.37)	-2.17 (1.41)
FTE employment after	21.17 (0.94)	21.03 (0.52)	-0.14 (1.08)	20.88 (1.01)	20.96 (0.76)	20.21 (1.03)	0.66 (1.44)	0.74 (1.27)
Change in mean FTE	-2.17 (1.22)	0.59 (0.47)	2.75 (1.31)	1.32 (0.81)	0.87 (0.67)	-2.04 (0.98)	3.36 (1.27)	2.91 (1.19)
Change, balanced sample	-2.28 (1.25)	0.47 (0.48)	2.75 (1.34)	1.20 (0.82)	0.71 (0.69)	-2.16 (1.01)	3.36 (1.30)	2.87 (1.22)
Change, temp. closed set to 0	-2.28 (1.25)	0.23 (0.49)	2.51 (1.35)	0.90 (0.87)	0.49 (0.69)	-2.39 (1.02)	3.29 (1.34)	2.88 (1.23)

NJ wage groups (wave-1 starting wage): $4.25 N=101, $4.26–$4.99 N=140, ≥$5.00 N=73. Row 3 is the unbalanced after – before difference; rows 4–5 use the store-level change on the balanced panel.

Table 2 — Means of key variables (NJ vs PA)

Variable	NJ	PA	t
Burger King (%)	41.09	44.30	-0.5
KFC (%)	20.54	15.19	1.2
Roy Rogers (%)	24.77	21.52	0.6
Wendy's (%)	13.60	18.99	-1.1
Company-owned (%)	34.14	35.44	-0.2
FTE employment, before	20.44	23.33	-2.0
% full-time employees	24.92	26.56	-0.6
Starting wage, before	4.61	4.63	-0.4
Wage = $4.25 (%)	30.51	32.91	-0.4
Price of full meal	3.35	3.04	4.0
Hours open, weekday	14.42	14.53	-0.3
Recruiting bonus (%)	23.56	29.11	-1.0
FTE employment, after	21.03	21.17	-0.1

Variable	NJ	PA	t
Starting wage, after	5.08	4.62	10.8
Price of full meal, after	3.41	3.03	5.1
Hours open, weekday, after	14.42	14.65	-0.6

Table 4 — Reduced-form models for the change in FTE employment

Sample: 365 stores; mean ΔFTE = -0.252, SD = 8.749. Dependent variable: change in FTE.

Model	NJ dummy	Wage gap	Controls	SER
(i)	+2.36 (1.15)	—	—	8.71
(ii)	+2.33 (1.15)	—	CO_OWNED, chain_2, chain_3, chain_4	8.70
(iii)	—	+15.89 (5.95)	—	8.68
(iv)	—	+15.19 (6.08)	CO_OWNED, chain_2, chain_3, chain_4	8.67
(v)	—	+12.03 (7.32)	CO_OWNED, chain_2, chain_3, chain_4, CENTRALJ, NORTHJ, PA1, PA2	8.67

Within-NJ treatment-intensity DiD (extension)

Sample restricted to New Jersey (all Pennsylvania observations dropped). No 0/1 control group; the continuous treatment intensity is `GAP = max(0, (5.05 − WAGE_ST)/WAGE_ST)`, so stores already paying ≥ $5.05 have GAP = 0 and act as the within-state reference. Dependent variable: change in FTE employment.

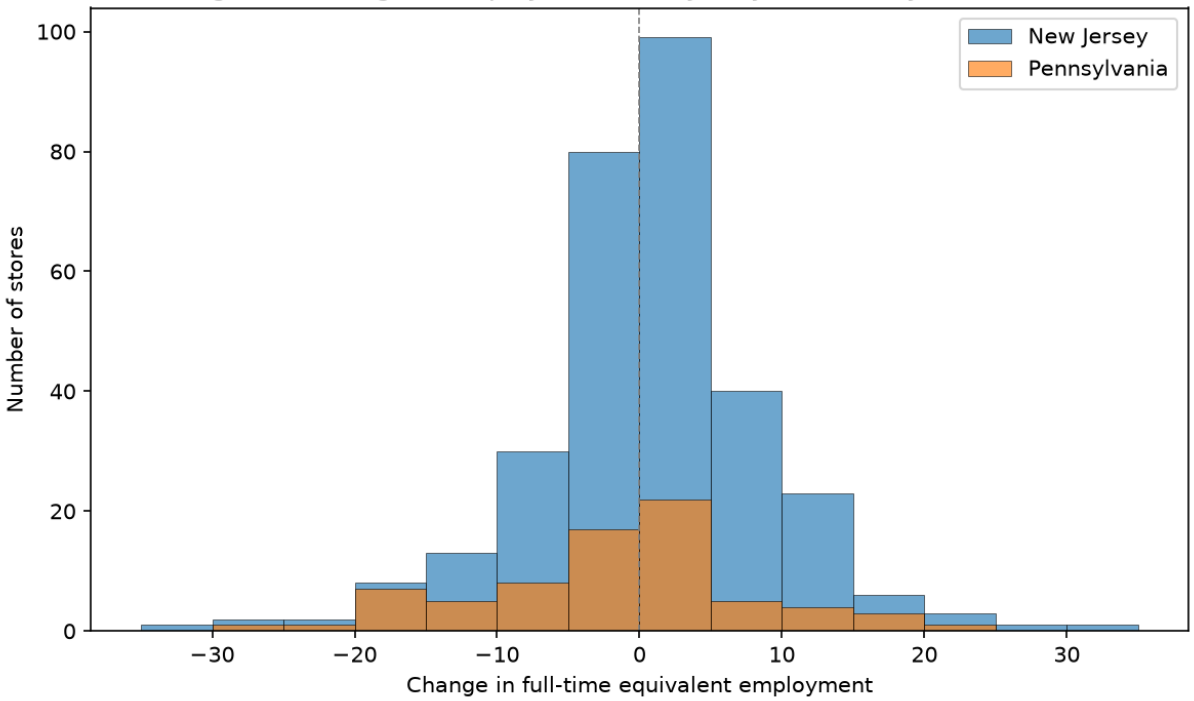
Regression sample: N = 293 NJ stores; mean ΔFTE = +0.213.

Specification	GAP coefficient	s.e.	t	p
GAP only	+13.08	6.60	1.98	0.048
+ chain & ownership	+12.27	6.82	1.80	0.073
+ region	+12.14	6.87	1.77	0.078

Discrete Low–High contrast (wage = $4.25 vs ≥ $5.00): **+3.36 FTE** (s.e. 1.28, p = 0.009); Mid–High = +2.87 (s.e. 1.20, p = 0.017). Both versions indicate no negative employment effect.

Figure 1 — Change in employment, New Jersey and Pennsylvania stores

Figure 1. Change in Employment, New Jersey and Pennsylvania Stores



Notes

- **FTE** = full-time employees + managers + 0.5 × part-time employees. Permanently-closed stores (STATUS2 = 3, six stores) are assigned zero wave-2 employment; temporarily-closed stores (STATUS2 ∈ {2,4,5}, four stores) are treated as missing, except in row 5 of Table 3 where they are set to zero.
- **The public-use `cardkrueger` file differs slightly from the paper's wave-1 staffing mix.** Wave-1 FTE and all wave-2 figures reproduce the published values exactly, but the file records fewer full-time and more part-time employees in wave 1, so the computed "% full-time employees" row (≈25% vs the paper's 32.8%) and the row-3 standard errors (a touch smaller) differ. The difference-in-differences point estimate and its conclusion are unaffected.
- **Headline result:** DiD = +2.75 FTE per store (SE 1.31), i.e. the New Jersey minimum-wage increase did not reduce fast-food employment.